1.

IP naslova nista enaka saj nista več na istem omrežju

Fizični PC vidi VM, a le v smeri "ven" (VM lahko dostopa do interneta preko PC-ja). Obratno pa fizični PC ne more kar tako "potrkati" na IP naslov VM-a, ker je ta skrit za NAT-om, razen če nastaviš posredovanje vrat (Port Forwarding).

VM dobi IP naslov iz istega nabora (subneta) kot tvoj fizični računalnik. Takrat sta napravi v istem lokalnem omrežju (LAN) in se lahko neposredno "pingata" ali komunicirata.

2.

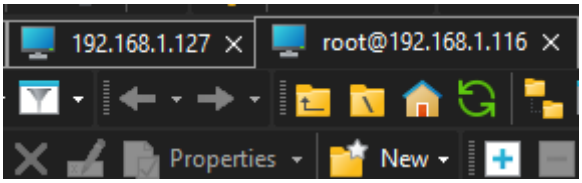
Spletna stran se prikaže

Preko orodja WinSCP lahko odpremo dve povezavi na VM in stem kopiramo datoteke preko 1. VM v 2. VM

```
root@nejcdolinar:~# systemctl start apache2
root@nejcdolinar:~# systemctl start ssh
root@nejcdolinar:~# systemctl enable apach2 ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Failed to enable unit: Unit apach2.service does not exist
root@nejcdolinar:~# systemctl enable apach2
Failed to enable unit: Unit apach2.service does not exist
root@nejcdolinar:~# systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
root@nejcdolinar:~# systectl status ssh
-bash: systectl: command not found
root@nejcdolinar:~# systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-26 11:44:07 CET; 59s ago
 Invocation: 4e658783f4d94d64b76692ecee9c3158
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 3716 (sshd)
      Tasks: 1 (limit: 506)
     Memory: 1.7M (peak: 2.2M)
        CPU: 27ms
     CGroup: /system.slice/ssh.service
             `-3716 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 26 11:44:07 nejcdolinar systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 26 11:44:07 nejcdolinar sshd[3716]: Server listening on 0.0.0.0 port 22.
Mar 26 11:44:07 nejcdolinar sshd[3716]: Server listening on :: port 22.
Mar 26 11:44:07 nejcdolinar systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
root@nejcdolinar:~# systemctl status apache2
* apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-26 11:44:11 CET; 1min 0s ago
 Invocation: cd5b4e3c0ead47c7aea6cc1058f8d73d
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 4059 (apache2)
      Tasks: 55 (limit: 506)
     Memory: 5.3M (peak: 5.6M)
        CPU: 48ms
     CGroup: /system.slice/apache2.service
             |-4059 /usr/sbin/apache2 -k start
             |-4060 /usr/sbin/apache2 -k start
             `-4079 /usr/sbin/apache2 -k start

Mar 26 11:44:11 nejcdolinar systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 26 11:44:11 nejcdolinar systemd[1]: Started apache2.service - The Apache HTTP Server.
root@nejcdolinar:~#
```



Not secure 192.168.1.127

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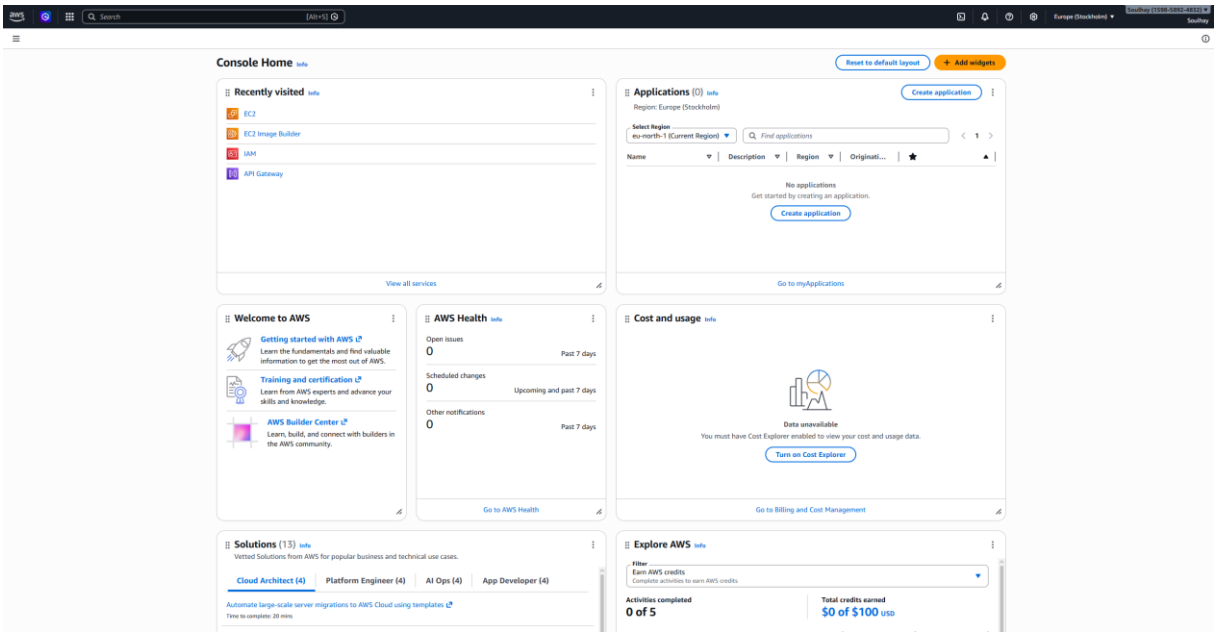
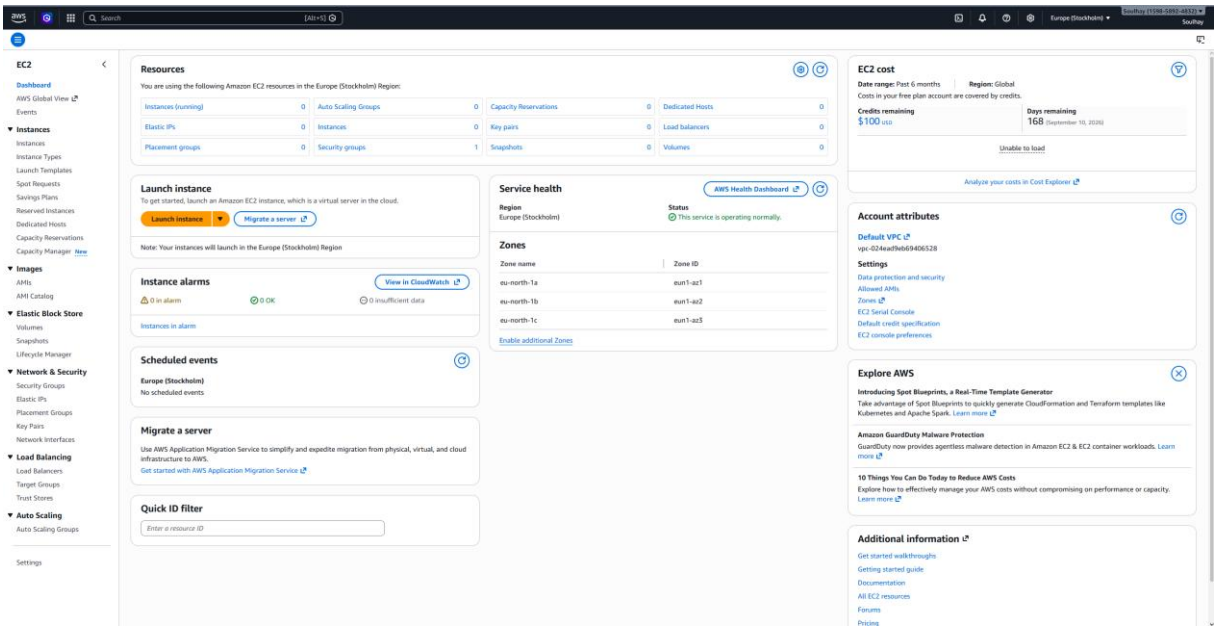
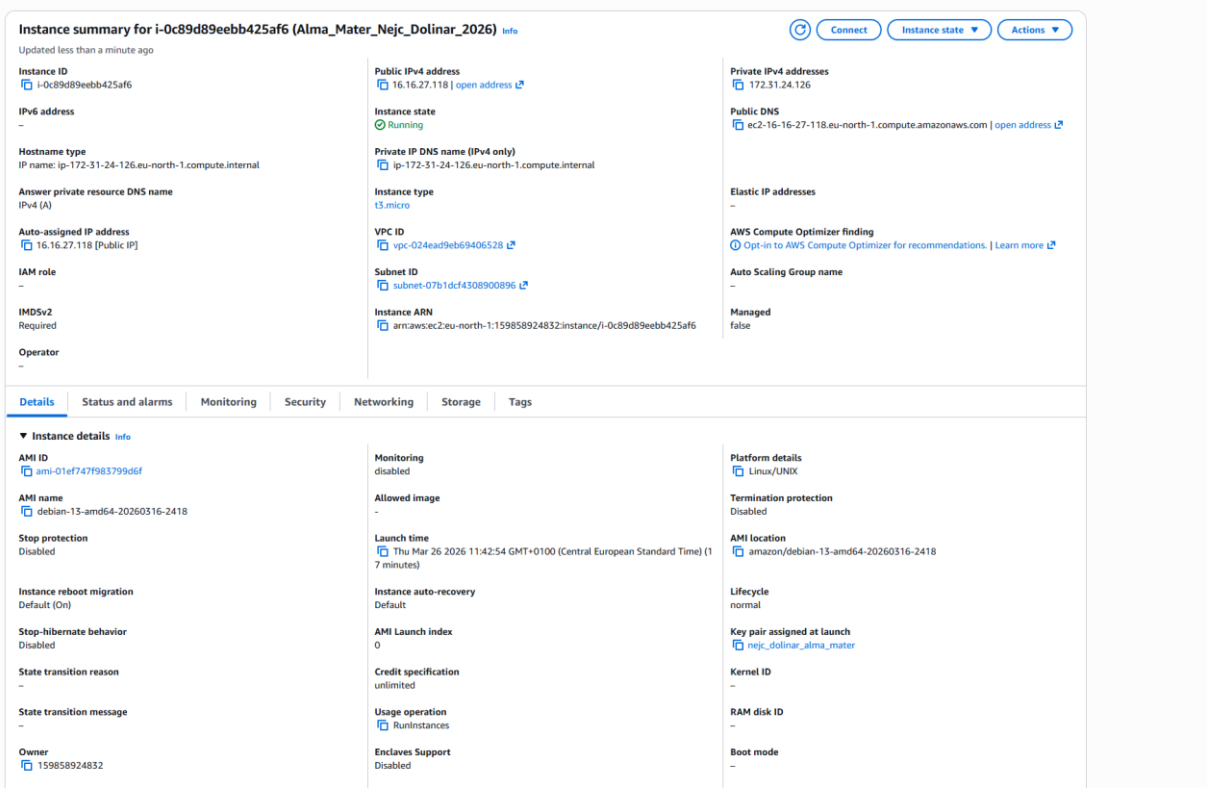
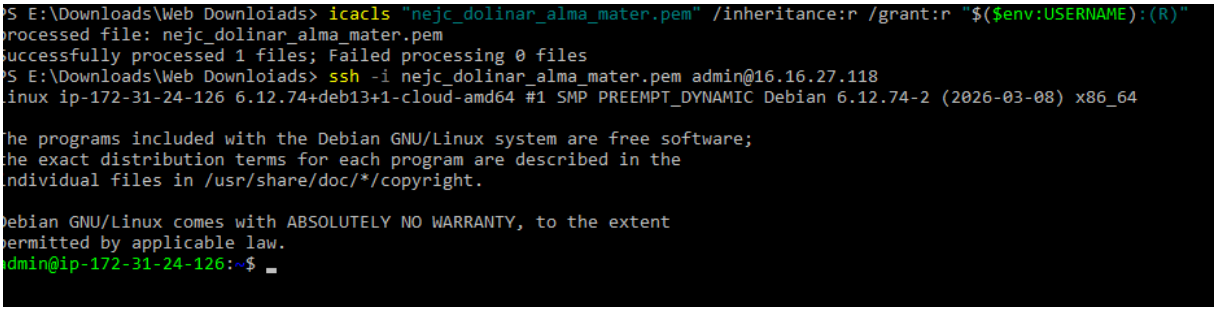
Nejc Dolinar




EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Resources
You are using the following Amazon EC2 resources in the Europe (Stockholm) Region:
Instances (running) 0
Auto Scaling Groups 0
Capacity Reservations 0
Dedicated Hosts 0
Elastic IPs 0
Instances 0
Key pairs 0
Load balancers 0
Placement groups 0
Security groups 1
Snapshots 0
Volumes 0
Launch instance
To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.
Launch instance
Migrate a server
Note: Your instances will launch in the Europe (Stockholm) Region
Instance alarms
View in CloudWatch
0 in alarm
0 OK
0 insufficient data
Instances in alarm
Scheduled events
Europe (Stockholm)
No scheduled events
Migrate a server
Use AWS Application Migration Service to simplify and expedite migration from physical, virtual, and cloud infrastructure to AWS.
Get started with AWS Application Migration Service
Quick ID filter
Enter a resource ID
Service health
AWS Health Dashboard
Region
Europe (Stockholm)
Status
This service is operating normally.
Zones
Zone name
Zone ID
eu-north-1a
eun1-az1
eu-north-1b
eun1-az2
eu-north-1c
eun1-az3
Enable additional Zones
EC2 cost
Date range: Past 6 months
Region: Global
Costs in your free plan account are covered by credits.
Credits remaining
$100 USD
Days remaining
168 (September 10, 2026)
Unable to load
Analyze your costs in Cost Explorer
Account attributes
Default VPC
vpc-024ead9eb69406528
Settings
Data protection and security
Allowed AMIs
Zones
EC2 Serial Console
Default credit specification
EC2 console preferences
Explore AWS
Introducing Spot Blueprints, a Real-Time Template Generator
Take advantage of Spot Blueprints to quickly generate CloudFormation and Terraform templates like Kubernetes and Apache Spark. Learn more
Amazon GuardDuty Malware Protection
GuardDuty now provides agentless malware detection in Amazon EC2 & EC2 container workloads. Learn more
10 Things You Can Do Today to Reduce AWS Costs
Explore how to effectively manage your AWS costs without compromising on performance or capacity. Learn more
Additional information
Get started walkthroughs
Getting started guide
Documentation
All EC2 resources
Forums
Pricing


Instance summary for i-0c89d89eebb425af6 (Alma_Mater_Nejc_Dolinar_2026) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0c89d89eebb425af6
Public IPv4 address
16.16.27.118 | open address
Private IPv4 addresses
172.31.24.126
IPv6 address
–
Instance state
Running
Public DNS
ec2-16-16-27-118.eu-north-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-24-126.eu-north-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-24-126.eu-north-1.compute.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
16.16.27.118 [Public IP]
VPC ID
vpc-024ead9eb69406528
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role
–
Subnet ID
subnet-07b1dcf4308900896
Auto Scaling Group name
–
IMDSv2
Required
Instance ARN
arn:aws:ec2:eu-north-1:159858924832:instance/i-0c89d89eebb425af6
Managed
false
Operator
–
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance details Info
AMI ID
ami-01ef747f983799d6f
Monitoring
disabled
Platform details
Linux/UNIX
AMI name
debian-13-amd64-20260316-2418
Allowed image
-
Termination protection
Disabled
Stop protection
Disabled
Launch time
Thu Mar 26 2026 11:42:54 GMT+0100 (Central European Standard Time) (17 minutes)
AMI location
amazon/debian-13-amd64-20260316-2418
Instance reboot migration
Default (On)
Instance auto-recovery
Default
Lifecycle
normal
Stop-hibernate behavior
Disabled
AMI Launch index
0
Key pair assigned at launch
nejc_dolinar_alma_mater
State transition reason
–
Credit specification
unlimited
Kernel ID
–
State transition message
–
Usage operation
RunInstances
RAM disk ID
–
Owner
159858924832
Enclaves Support
Disabled
Boot mode
–


S E:\Downloads\Web Downloads> icacls "nejc_dolinar_alma_mater.pem" /inheritance:r /grant:r "$($env:USERNAME):(R)"
rocessed file: nejc_dolinar_alma_mater.pem
uccessfully processed 1 files; Failed processing 0 files
S E:\Downloads\Web Downloads> ssh -i nejc_dolinar_alma_mater.pem admin@16.16.27.118
inux ip-172-31-24-126 6.12.74+deb13+1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64
he programs included with the Debian GNU/Linux system are free software;
he exact distribution terms for each program are described in the
ndividual files in /usr/share/doc/*/copyright.
ebian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
ermitted by applicable law.
dmin@ip-172-31-24-126:~$

```
 package can be upgraded. Run 'apt list --upgradable' to see it.
Upgrading:
  libfreetype6

Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 0
  Download size: 452 kB
  Space needed: 1024 B / 6733 MB available

Get:1 file:/etc/apt/mirrors/debian-security.list Mirrorlist [47 B]
Get:2 https://cdn-aws.deb.debian.org/debian-security trixie-security/main amd64 libfreetype6 amd64 2.13.3+dfsg-1+deb13u1 [452 kB]
Fetched 452 kB in 0s (6275 kB/s)
Apt-listchanges: Reading changelogs...
(Reading database ... 30202 files and directories currently installed.)
Preparing to unpack .../libfreetype6_2.13.3+dfsg-1+deb13u1_amd64.deb ...
Unpacking libfreetype6:amd64 (2.13.3+dfsg-1+deb13u1) over (2.13.3+dfsg-1) ...
Setting up libfreetype6:amd64 (2.13.3+dfsg-1+deb13u1) ...
Processing triggers for libc-bin (2.41-12+deb13u2) ...
admin@ip-172-31-24-126:~$ sudo ufw allow OpenSSH
sudo: ufw: command not found
admin@ip-172-31-24-126:~$ sudo apt install -y ufw
Installing:
  ufw

Suggested packages:
  rsyslog

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 169 kB
  Space needed: 880 kB / 6733 MB available

Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [38 B]
Get:2 https://cdn-aws.deb.debian.org/debian trixie/main amd64 ufw all 0.36.2-9 [169 kB]
Fetched 169 kB in 0s (1990 kB/s)
Preconfiguring packages ...
Selecting previously unselected package ufw.
(Reading database ... 30202 files and directories currently installed.)
Preparing to unpack .../archives/ufw_0.36.2-9_all.deb ...
Unpacking ufw (0.36.2-9) ...
Setting up ufw (0.36.2-9) ...
Creating config file /etc/ufw/before.rules with new version
Creating config file /etc/ufw/before6.rules with new version
Creating config file /etc/ufw/after.rules with new version
Creating config file /etc/ufw/after6.rules with new version
Created symlink '/etc/systemd/system/multi-user.target.wants/ufw.service' → '/usr/lib/systemd/system/ufw.service'.
Processing triggers for man-db (2.13.1-1) ...
admin@ip-172-31-24-126:~$ sudo ufw allow OpenSSH
Rules updated
Rules updated (v6)
admin@ip-172-31-24-126:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
admin@ip-172-31-24-126:~$
```

```
admin@ip-172-31-24-126:~$ sudo apt install -y apache2
```

```
admin@ip-172-31-24-126: $ sudo systemctl start apache2
admin@ip-172-31-24-126: $ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
admin@ip-172-31-24-126: $ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-26 11:12:58 UTC; 2min 0s ago
 Invocation: 380f2ac45ea849289c5cc71b6f576cec
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2334 (apache2)
      Tasks: 55 (limit: 1097)
     Memory: 5.3M (peak: 5.6M)
        CPU: 50ms
     CGroup: /system.slice/apache2.service
             ├─2334 /usr/sbin/apache2 -k start
             ├─2335 /usr/sbin/apache2 -k start
             └─2336 /usr/sbin/apache2 -k start

Mar 26 11:12:58 ip-172-31-24-126 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 26 11:12:58 ip-172-31-24-126 apachectl[2333]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using fe80::4a7:3ff:fe4a:7e29%ens5. Set the 'ServerName' directive globally to suppress this message
Mar 26 11:12:58 ip-172-31-24-126 systemd[1]: Started apache2.service - The Apache HTTP Server.
admin@ip-172-31-24-126: $
```

sg-017cd8ac6849ddbaf - launch-wizard-1

Actions

Details

Security group name	Security group ID	Description	VPC ID
launch-wizard-1	sg-017cd8ac6849ddbaf	launch-wizard-1 created 2026-03-26T10:41:18.860Z	vpc-024ead9eb69406528
Owner	**Inbound rules count**	**Outbound rules count**	
159858924832	1 Permission entry	1 Permission entry	

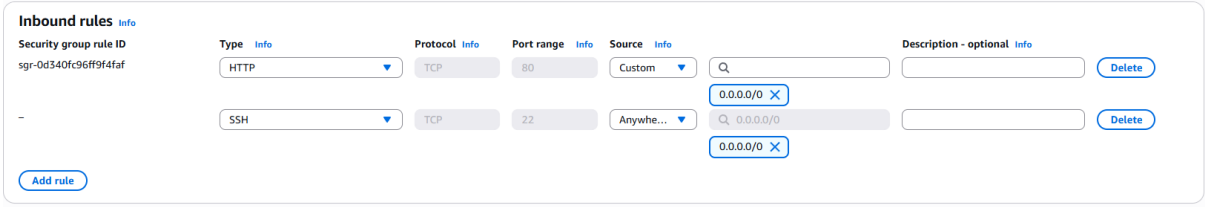
Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

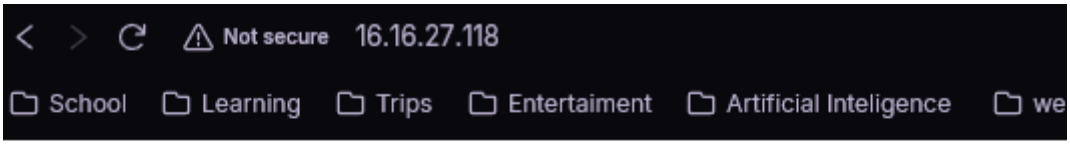
Inbound rules (1)

Manage tags | Edit inbound rules

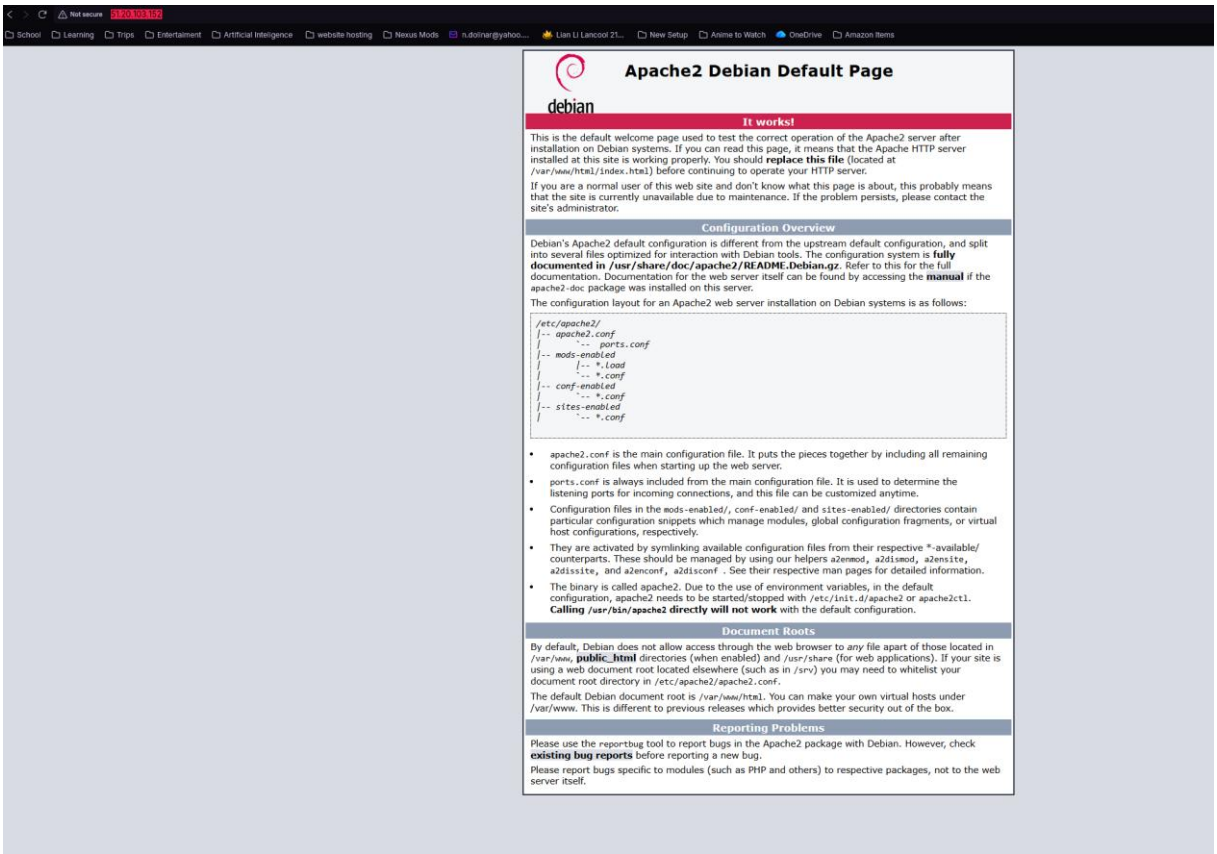
Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0d340fc96ff9f4faf	IPv4	HTTP	TCP	80	0.0.0.0/0	-



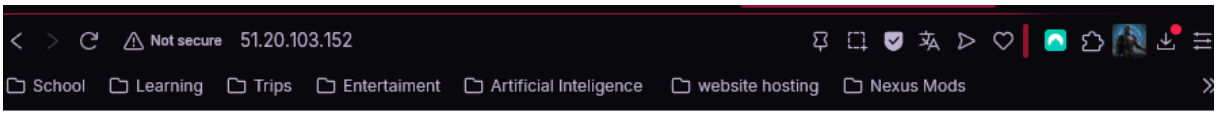


Pozdrav iz Instance 1!




```
PS E:\Downloads\Web Downloiads> scp -i "E:\Downloads\Web Downloiads\nejc_dolinar_alma_mater.pem" admin@16.16.27.118:/var/www/html/index.html "E:\Downloads\Web Downloiads\index.html"
```

```
PS E:\Downloads\Web Downloiads> scp -i "E:\Downloads\Web Downloiads\nejc_dolinar_alma_mater.pem" "E:\Downloads\Web Downloiads\index.html" admin@51.20.103.152:/tmp/index.html
```



Pozdrav iz Instance 1!

Kakšna je razlika med NAT in Bridged omrežjem?

- **NAT** (Network Address Translation) — VM deli IP naslov gostitelja. Zunanji svet vidi samo IP gostitelja, ne VM-ja. VM ima dostop do interneta, a do njega od zunaj ni mogoče dostopati neposredno.
- **Bridged** — VM dobi lasten IP naslov v istem omrežju kot gostitelj. Deluje kot samostojna naprava v omrežju, dostopna od zunaj.

Zakaj v NAT načinu dostop pogosto ne deluje?

Ker VM nima javnega IP naslova — skrit je za IP gostitelja. Ko npr. odpremo brskalnik in vpišemo IP VM-ja, zahteva ne pride do VM-ja, ker router ne ve, da jo mora posredovati nanj. Potrebno je ročno nastavljanje **port forwardinga**.

Kakšna je razlika med javnim in zasebnim IP naslovom?

- **Zasebni IP** (npr. 192.168.x.x, 172.31.x.x) — uporablja se znotraj lokalnega omrežja, ni dostopen iz interneta.
- **Javni IP** (npr. 16.16.27.118) — edinstveni naslov na internetu, prek katerega je strežnik dostopen od kjerkoli na svetu.

Kakšna je vloga SSH?

SSH (Secure Shell) omogoča **varno oddaljeno upravljanje** strežnika prek šifriranega tunela. Namesto fizičnega dostopa do strežnika se povežemo prek terminala in izvajamo ukaze, kot da bi sedeli pred njim.

Kakšna je vloga Apache strežnika?

Apache je **spletni strežnik** — sprejema HTTP zahteve od brskalnikov in jim vrača spletne strani (HTML, slike, itd.). Brez njega strežnik ne more servirati spletnih vsebin.

Kaj omogoča Security Group v AWS?

Security Group je **požarni zid** na ravni AWS — določa kateri promet sme priti do instance in kateri ne. Npr. brez pravila za port 80 Apache teče, a ga nihče od zunaj ne more doseči. Pravila nastavljamo po tipu (SSH, HTTP), portu in viru IP.